

PRODUCT REQUIREMENTS DOCUMENT

V1.0.0 APPROVED

SkillPulse

Daily Execution, Learning Curve & Upskill Tracker with Proof-of-Work Screen Capture and Real-Time Growth Analytics

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TARGET RELEASE

Q3 2026 (Production Release)

PRODUCT STATUS

Complete / Live Deployment

REPOSITORY URI

github.com/manukotari/SkillPulse

1. Executive Summary & Vision

1.1 Problem Statement

Software developers, engineering students, and upskilling professionals consistently face the "**Daily Drift Dilemma**":

- **Disconnection between Day-to-Day Work and Career Upskilling:** Developers complete tasks every day, yet fail to bridge the skills gap needed for promotions or senior architectural roles.
- **Lost Daily Learnings:** Tough bugs solved, mental models discovered, and "Aha!" insights vanish within 48 hours because there is no frictionless reflection mechanism.
- **Absence of Tangible Proof of Work:** Portfolios and resumes claim skills, but lack verifiable, timestamped visual evidence of hands-on execution (e.g. profiler traces, database query plans, passed assertions).
- **Privacy Concerns:** Many developers are reluctant to log company or personal work details on cloud SaaS apps that scrape or monetize proprietary data.

1.2 Solution Overview

SkillPulse is an aesthetic, **Local-First Daily Command Center** that bridges daily execution with intentional career upskilling. It combines:

1. Daily Task Management

EXECUTION

Domain-categorized checklist with priority tags, estimated vs. actual minutes, and one-click time loggers.

2. Screen Capture Proof-of-Work

VERIFICATION

Native browser screen-access popup capturing high-res snapshots of IDEs/terminals with pen/highlighter markup.

3. Daily Learning Curve Journal

RETENTION

Capture "Aha!" breakthroughs, solutions to blockers, and comprehension scores (1-10) to map learning velocity.

4. Upskill Engine ("Changes Required")

PROGRESSION

Pinpoints exact architectural and engineering shifts required to reach target seniority benchmarks.

1.3 Target Personas

PERSONA	CORE PAIN POINTS	PRIMARY VALUE DELIVERED
Mid-Level Software Engineer	Wants promotion to Senior/Lead; struggles to identify concrete technical gaps (e.g., distributed caching, DB partitioning).	Clear "Changes Required to Upskill" action plans, skill gap radar, and milestone tracking.
Self-Taught Developer / Learner	Overwhelmed by tutorials; loses track of what they actually mastered vs. passive reading.	Daily 1-10 mastery curve rating and learning reflection journal that cements retention.
Technical Interview Candidate	Needs proof of practical problem-solving beyond LeetCode statistics.	Visual gallery of screen captures verifying profiling, architectural diagrams, and benchmark runs.

Core Product Philosophy

SkillPulse treats personal growth with the same architectural rigor as enterprise software: measured daily, backed by proof, and tracked against explicit engineering benchmarks.

2. Goals, Scope & User Journey

2.1 Goals & Success Metrics (v1.0)

100% LOCAL-FIRST PRIVACY	<50ms UI INTERACTION LATENCY	0 KB SERVER TELEMETRY / LEAK	30 Days HABIT HEATMAP TRACKING
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Key Performance Indicators (KPIs):

- **Daily Active Streak Retention:** Percentage of users maintaining a ≥ 5 -day streak of logging either a task or learning reflection.
- **Screen Capture Attachment Rate:** Average number of verified visual snapshots attached per completed high-priority task (Target: ≥ 0.8 proofs/task).
- **Upskill Gap Closure Velocity:** Time taken to advance a tracked skill from *"Identified"* to *"Mastered"* (Target: 3–6 weeks).

2.2 Out-of-Scope (Non-Goals for v1.0)

- **Centralized Multi-Tenant SaaS:** v1.0 intentionally avoids remote database backends to ensure zero infrastructure cost and maximum user data privacy.
- **Automated Time Trackers / Activity Surveillance:** SkillPulse does not secretly log keystrokes or record screen video streams in the background; captures are initiated on-demand by the user.
- **Team / Enterprise RBAC:** v1.0 focuses exclusively on the individual developer's personal cockpit.

2.3 Day-in-the-Life User Journey

TIME OF DAY	USER ACTION	SYSTEM BEHAVIOR	ARTIFACT CREATED
09:00 AM (Morning)	Opens SkillPulse, reviews yesterday's streak, adds 3 daily tasks with priority and domains.	Updates today's completion progress bar on dashboard; initializes timers.	Daily Task list
01:30 PM (Midday)	Resolves PostgreSQL query bottleneck; clicks <i>"Capture Screen"</i> popup and snaps pgAdmin EXPLAIN trace.	Renders canvas snapshot; user annotates latency drop with neon highlighter; links to Task-4.	ScreenEvidence in IndexedDB

TIME OF DAY	USER ACTION	SYSTEM BEHAVIOR	ARTIFACT CREATED
05:30 PM (Evening)	Clicks " <i>+ Log Today's Learning</i> " in Learning Journal; rates mastery at 9/10 and writes "Aha!" insight.	Recalculates 7-day average mastery index; plots new data point on Learning Curve area chart.	LearningEntry
06:00 PM (Wrap-up)	Opens Upskill Roadmap; checks off completed milestone on " <i>Distributed Systems</i> "; inspects Radar.	Progress bar updates to 67%; radar updates geometry; heatmap illuminates today's green cell.	Upskill milestone check

3. Functional Feature Specifications

3.1 Module FR-1: Daily Task Management Hub

The Daily Task Hub serves as the execution engine for organizing daily work into structured domain milestones.

REQ ID	FEATURE	FUNCTIONAL DESCRIPTION	PRIORITY
FR-1.1	Domain Categorization	Every task must belong to one of 7 standardized engineering domains: <i>Frontend, Backend, System Design, DevOps, Algorithms, Architecture, Soft Skills</i> .	PO - MUST
FR-1.2	Priority Matrix	Tasks support <code>high</code> , <code>medium</code> , <code>low</code> tags with neon-coded visual badges.	PO - MUST
FR-1.3	3-State Progression	Tasks transition smoothly across <code>todo</code> → <code>in-progress</code> → <code>completed</code> via one-click status toggle.	PO - MUST
FR-1.4	Time Tracking & Quick-Log	Supports estimated minutes vs actual minutes with a dedicated <code>+15m</code> quick button on each card for in-flow logging.	PO - MUST
FR-1.5	Evidence Linking	Tasks display a counter badge showing how many screen captures are attached, with one-click direct evidence preview.	P1 - HIGH

3.2 Module FR-2: Screen Access & Proof-of-Work Capture Popup

A core differentiator of SkillPulse is enabling users to capture verifiable visual snapshots of their daily work.

REQ ID	FEATURE	FUNCTIONAL DESCRIPTION	PRIORITY
FR-2.1	Native Screen Access	Invokes <code>navigator.mediaDevices.getDisplayMedia()</code> to allow selecting full screen, application window, or browser tab.	PO - MUST
FR-2.2	Instant Canvas Freeze	Renders the video track frame onto an offscreen canvas, extracts a PNG data URL, and terminates media tracks to release hardware.	PO - MUST

REQ ID	FEATURE	FUNCTIONAL DESCRIPTION	PRIORITY
FR-2.3	Interactive Markup Suite	Provides pen tool (3px stroke), highlighter tool (18px translucent stroke), 5 neon color pickers, and undo/retake actions.	P0 - MUST
FR-2.4	IndexedDB Media Storage	Persists full-resolution snapshots into browser IndexedDB (SkillPulseDB) avoiding LocalStorage 5MB limits.	P0 - MUST
FR-2.5	Clipboard & Upload Fallback	Listens for Ctrl+V clipboard paste of screenshots, supports file upload, and provides a mock test canvas generator.	P1 - HIGH

3.3 Module FR-3: Daily Learning Curve & Reflection Journal

The Neural Retention Loop

Passive reading produces ~10% retention. Active logging of the "Aha!" moment and error resolution produces >75% concept retention.

REQ ID	FEATURE	FUNCTIONAL DESCRIPTION	PRIORITY
FR-3.1	1-10 Mastery Slider	User rates daily comprehension from 1 (Exposed) to 10 (Intuitive mastery / Can teach). Feeds directly into the analytics curve.	P0 - MUST
FR-3.2	The "Aha!" Breakthrough	Dedicated input box capturing the pivotal concept that clicked during that day's session.	P0 - MUST
FR-3.3	Obstacle & Solution Pair	Structured pair documenting the specific blocker (e.g. race condition) and its resolution (e.g. atomic Redis lock).	P0 - MUST
FR-3.4	Domain & Tag System	Allows custom hashtags (#React19 , #PostgreSQL , #Kafka) with domain categorization.	P1 - HIGH

3.4 Module FR-4: Upskill Engine ("Changes Required to Upskill")

The Upskill Engine transforms vague desires (e.g., *"I want to become a Senior Engineer"*) into precise, measurable architectural changes.

REQ ID	FEATURE	FUNCTIONAL DESCRIPTION	PRIORITY
FR-4.1	Proficiency Delta Meter	Compares Current Level (1-10) vs Target Level (1-10) with a visual dual-color progress bar highlighting the gap delta.	PO - MUST
FR-4.2	"Changes Required" Box	Explicit bulleted list of concrete engineering changes required (e.g., <i>"Master Write-Ahead Logging"</i> , <i>"Implement Circuit Breaker pattern"</i>).	PO - MUST
FR-4.3	Milestone Action Checklist	Interactive checklist of project steps; automatically calculates completion percentage and advances status upon full completion.	PO - MUST
FR-4.4	Target Deadlines	Sets target completion dates and tracks status: identified → in-progress → mastered .	P1 - HIGH

3.5 Module FR-5: Real-Time Analytics Dashboard

The Analytics Dashboard provides instant visual feedback across productivity, comprehension curves, and consistency.

Learning Curve Dynamics

RECHARTS AREA

Dual-axis chart plotting **Mastery Score (Area gradient)** alongside **Hours Invested (Dashed line)** over time, with tooltips showing topics and breakthroughs.

Upskill Gap Radar

RECHARTS RADAR

Multi-axis radar comparing current proficiency vs target benchmark level across up to 6 competencies simultaneously.

Domain Focus Donut

RECHARTS PIE

Interactive donut chart showing task distribution across disciplines (Frontend, Backend, System Design, DevOps, etc.).

28-Day Habit Heatmap

GITHUB-STYLE

Visual activity grid tracking daily execution density. Clicking any square filters all tasks and learnings for that day.

3.6 Module FR-6: Data Portability & Backup Engine

- **JSON Backup Export:** Serializes tasks, learning logs, upskill roadmaps, and evidence into a structured JSON file for backup.
- **JSON Import:** Parses and restores complete data state across browsers or machines.
- **Demo Data Reset:** Instantly resets application state with realistic preloaded engineering mock data for demonstrations.

3.7 Module FR-7: Interactive User Onboarding Tour

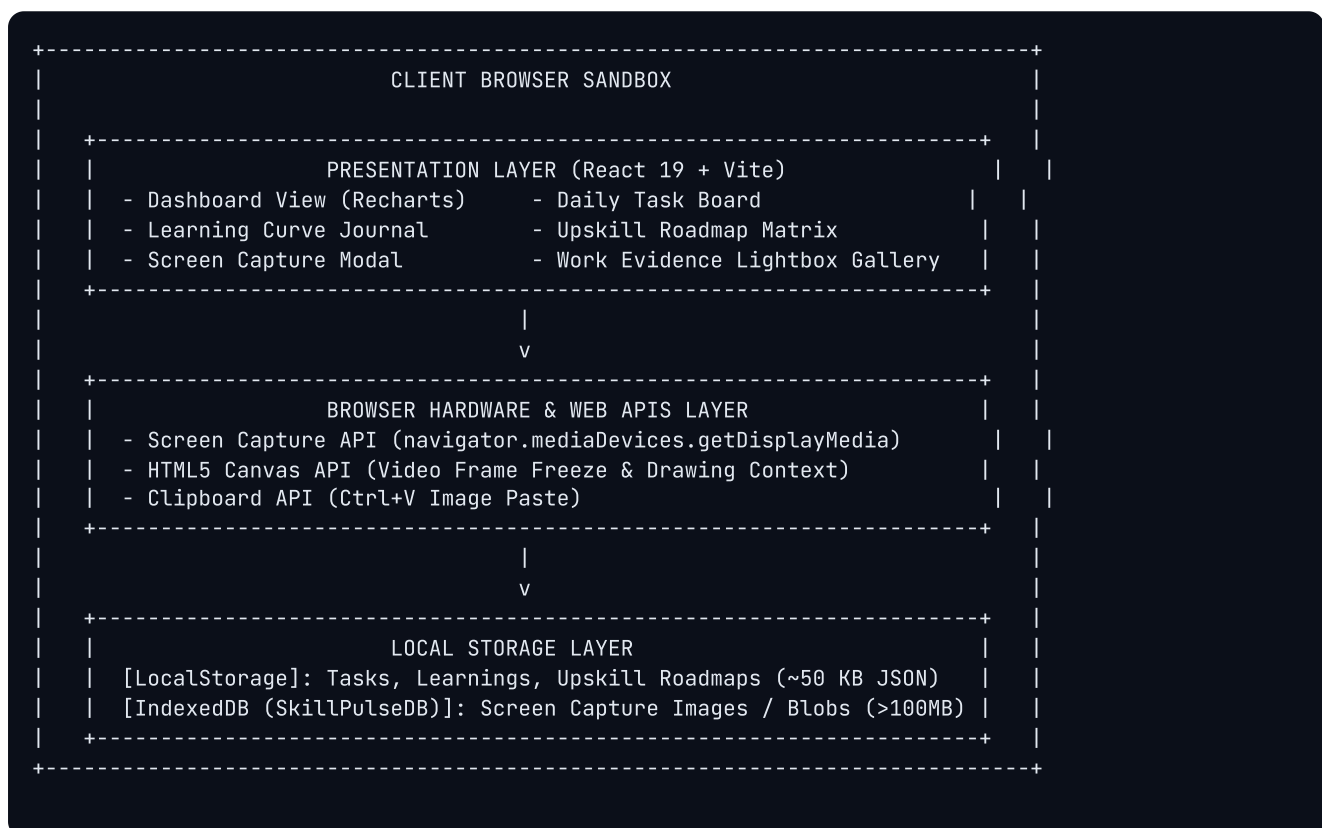
- 5-step guided modal accessible via the "**User Guide**" button in the navigation header.
- Walks first-time users through the full morning-to-evening workflow.

4. Technical Architecture & Data Models

4.1 Architecture Diagram

Client-Side Local-First Architecture

SkillPulse runs entirely inside the client sandbox. It has **zero external server dependencies**, rendering it completely private, lightning-fast, and immune to downtime.



4.2 Core TypeScript Data Entities

All core entities are strongly typed in `src/types/index.ts`:

```

// 1. Task Entity
interface Task {
  id: string;
  title: string;
  description?: string;
  category: 'Frontend' | 'Backend' | 'System Design' | 'DevOps' | 'Algorithms' | 'Architecture' | 'So
  priority: 'high' | 'medium' | 'low';
  status: 'todo' | 'in-progress' | 'completed';
  date: string; // YYYY-MM-DD
  estimatedMinutes: number;
  actualMinutes: number;
  evidenceIds: string[]; // Foreign keys to ScreenEvidence
}
  
```

```

    createdAt: string;
}

// 2. Learning Reflection Entity
interface LearningEntry {
  id: string;
  date: string;
  topic: string;
  category: TaskCategory;
  summary: string;
  ahaMoment: string;
  challengeFaced: string;
  howSolved: string;
  masteryScore: number; // 1 to 10
  hoursSpent: number;
  tags: string[];
  evidenceIds: string[];
}

// 3. Upskill Gap & Changes Required Entity
interface UpskillGap {
  id: string;
  skillName: string;
  category: TaskCategory;
  currentProficiency: number; // 1 - 10
  targetProficiency: number; // 1 - 10
  reasonForGap: string;
  changesRequired: string[]; // Concrete engineering changes
  actionItems: { id: string; text: string; completed: boolean }[];
  priority: 'high' | 'medium' | 'low';
  status: 'identified' | 'in-progress' | 'mastered';
  targetDate: string;
}

// 4. Screen Capture Evidence Entity
interface ScreenEvidence {
  id: string;
  timestamp: string;
  title: string;
  notes: string;
  imageData: string; // Data URL PNG stored in IndexedDB
  taskId?: string;
  category?: TaskCategory;
}

```

5. Non-Functional Requirements & Design System

5.1 Non-Functional Requirements (NFRs)

CATEGORY	SPECIFICATION	VERIFICATION METHOD
Performance	Initial page load under 1.0s; chart render under 50ms; zero layout shift (CLS = 0.00).	Lighthouse Audit & Chrome DevTools Profiler
Security & Privacy	100% data residency inside client sandbox. No telemetry or image uploads to third-party endpoints.	Network Tab inspection (0 external POST calls)
Reliability	Works completely offline without network access. Graceful degradation when media capture is rejected.	Offline Mode Service Simulation & Fallbacks
Compatibility	Chromium (Chrome 90+, Edge 90+, Brave), Safari 16+, Firefox 110+.	Cross-browser automated & manual checks

5.2 Design System & Visual Identity

The application implements an executive **Cyberpunk Glassmorphism** palette:

TOKEN	HEX CODE	VISUAL REPRESENTATION	APPLICATION PURPOSE
Canvas Background	#0B0F19	 Deep Midnight	Main view backdrop and contrast foundation
Primary Accent	#6366F1	 Electric Indigo	Brand buttons, active tab indicators, task badges
Velocity Accent	#06B6D4	 Luminous Cyan	Learning curve area gradient, profiler tags
Success Accent	#10B981	 Neon Emerald	Completed tasks, verified evidence badges, mastery index
Warning Accent	#F59E0B	 Amber Glow	"Aha!" breakthrough cards, medium priority badges
Critical Accent	#F43F5E	 Rose Flare	High-priority tags, obstacle alert blocks

5.3 Future Roadmap (v2.0 & Beyond)

MILESTONE	KEY DELIVERABLE	STRATEGIC VALUE
v2.0 (Q4 2026)	AI Upskilling Copilot (Gemini API)	Analyzes daily reflections and automatically recommends customized learning modules & papers.
v2.1 (Q1 2027)	Multi-Device Cloud Sync	Optional Supabase / Firebase encrypted cloud sync for multi-workstation engineers.
v2.2 (Q2 2027)	Public Portfolio Exporter	Generates a shareable, static "Proof of Work" webpage showcasing verified screenshots.

Document Sign-off

SkillPulse Product Requirements Document (PRD) v1.0.0 is finalized and verified against production deployment at <http://localhost:5173/> and GitHub repository [manukotari/SkillPulse](https://github.com/manukotari/SkillPulse).